

This practice matches the real quiz question for question. Work every problem, then check the worked key. Anything you miss is exactly what to study.

Vocabulary: Write the term from the word bank that matches each description. Not every term is used. (1 Point Each)

numerical expression algebraic expression variable term constant coefficient
evaluate

1. An expression containing at least one variable.

2. A letter used to represent an unknown or changing value. _____

3. A term that is a number only – its value never changes. _____

4. To find the value of an expression by carrying out all of its operations.

Problems: (2 Points Each)

5. Write a numerical expression for the verbal expression. **DO NOT EVALUATE.**
seven times the difference of fifteen and eight

6. Write an algebraic expression for the verbal expression:
nine less than the quotient of k and 3

7. Evaluate: $40 - 6 + 2^4 - 7 \cdot 5$

8. Evaluate: $(5^3 - 5) / (9 + 6)$

9. Evaluate $a^2(3b - a + 2) \div c$, if $a = 4$, $b = 3$, and $c = 8$.

10. Error Analysis – Gus evaluated $4x^2$ when $x = 3$ and wrote 144. Identify his mistake, then evaluate correctly.

11. Identify the terms, the coefficients, and the constant in $7x^2 + 2x - 5$

12. Maya works at a café. She earns \$12 per hour making drinks and \$9 per hour at the register. She spends \$5 on lunch each day she works.

a. Write an algebraic expression for Maya's profit if she works d hours making drinks and r hours at the register.

b. Evaluate your expression to find her profit on a day she makes drinks for 4 hours and works the register for 3 hours.